Sahil Shaik

858-663-7728 | sahil.r.shaik@gmail.com | linkedin.com/in/sahil-shaik07 | Portfolio: Attached | U.S. Citizen

Education

Georgia Institute Of Technology **2025 - 2028**

Bachelor of Science in Electrical Engineering - GPA: 3.72

Work Experience

Data Science and Systems Informatics Lab @ UGA | Athens, GA **Aug 2025 – May 2026**

Undergraduate Research Assistant *– Lab researching potential hardware and software solutions for industrial problems*

- Developed wiring and assembled hardware for both a robotic hand and associated glove to enable real-time gesture mapping, integrating 4 flex sensors in voltage divider configurations to capture as servo position inputs
- Calibrated resistance thresholds per flex-sensor through empirical testing, translating raw finger bend range into accurate, proportional servo angle values by placing a known voltage under a voltage divider setup, developing a watchdog program monitoring position, speed, torque, and acceleration
- Interfaced with register-based SCS009 serial bus servos via a half-duplex TTL adapter, multiplexing TX and RX over a single line to transmit commands and enabling the robotic hand to mimic glove gestures

System Technology Works LLC | Peachtree Corners, GA **May 2025 – Aug 2025**

Electrical Engineering Intern *– Startup, developing entertainment robots for clients including Marvel and Paramount Pictures*

- Implemented a ROS-based multithreaded control system for an 8-legged robot, coordinating leg movement across threads to resolve synchronization issues and achieve measurably smoother locomotion
- Integrated and configured onboard sensors including microphone and speaker peripherals on the Jetson Nano, coordinating sensor I/O with the robot's control software to support real-time interactive behavior
- Developed interrupt-driven fault detection firmware for the motor control system of an entertainment robot on the NVIDIA Jetson Nano, configuring an ISR to monitor digital fault signals from the gate drivers and disabling PWM motor drive output within microseconds of a fault condition, preventing potential motor and gate driver damage during live operation

Projects and Activities

Wireless Sensor Hub **July 2026**

- Developed a wireless sensor hub on the STM32 (ARM Cortex-M4) running FreeRTOS, architecting a multi-task pipeline with dedicated tasks for IMU acquisition, UART transmission, and LED status indication, utilizing message queues for inter-task communication and deliberate stack allocation to operate within the 20KB RAM constraint
- To avoid library dependencies and demonstrate protocol-level understanding, implemented SHTP (Sensor Hub Transport Protocol) manually over I2C without external libraries, handling the full communication sequence including reset control, advertisement packet draining, set feature commands, and data report to parse and transmit data across the accelerometer, gyroscope, and magnetometer within the BNO085
- Transmitted structured sensor data over UART to an ESP32 bridge, which published readings to an MQTT broker over Wi-Fi with OTA firmware update support, enabling wireless firmware deployment without physical access to the device

MicroMouse Club | Athens, GA **Aug 2025 – May 2026**

National Competition Team *- Club developing an autonomous maze-solving robot for national competition*

- Jointly implemented competition board firmware, integrating motor control, PID, FloodFill navigation, IR sensor data acquisition, and BNO085 IMU and HC-06 Bluetooth modules over I2C and UART, producing a competition-ready autonomous maze-solving robot on the STM32 target hardware
- Identified an ADC channel conflict arising from multiple sensors competing across a single ADC peripheral, implementing a DMA circular scan conversion to continuously cycle through channels and write results directly to memory for real-time firmware access without CPU intervention
- Designed and implemented a FloodFill maze-solving algorithm from scratch, dynamically updating cell distance values as new walls were discovered during navigation to continuously optimize the robot's path to the maze center

Skills

Embedded Software: RTOS, I2C, UART, SPI, ADC, DAC, CAN, PWM, DMA, Timers, C, C++, Python, Java
Embedded Hardware: ARM Cortex, PCB Design, Schematic Design, Oscilloscopes, Digital Multimeters, Logic Analyzers
Programs: Altium Designer, MATLAB, EAGLE, SolidWorks, VS Code

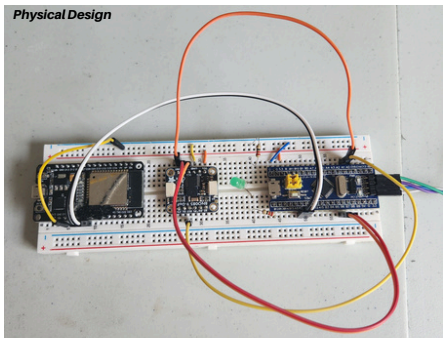
SAHIL SHAIK

ELECTRICAL ENGINEERING AT THE UNIVERSITY OF GEORGIA

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WIRELESS SENSOR CONTROL HUB

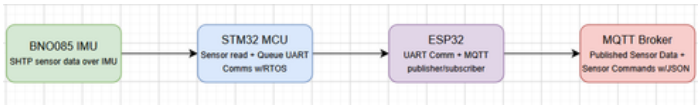
Physical Design



Situation

- Needed a **real-time embedded platform** to continuously acquire, process and wirelessly transmit multi-axis inertial and magnetic sensor data
- Required a **memory-constrained RTOS implementation** operating within the STM32 Blue Pill's 20KB RAM while maintaining concurrent sensor acquisition and communication tasks

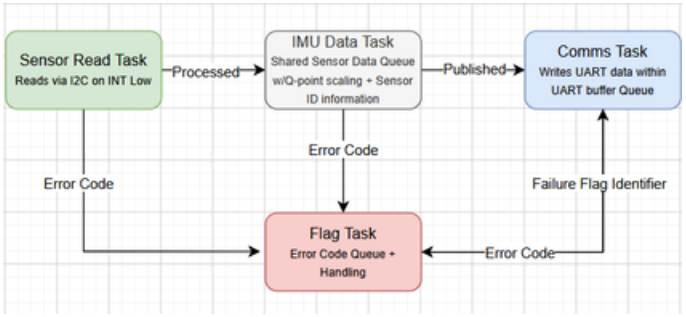
System Pipeline



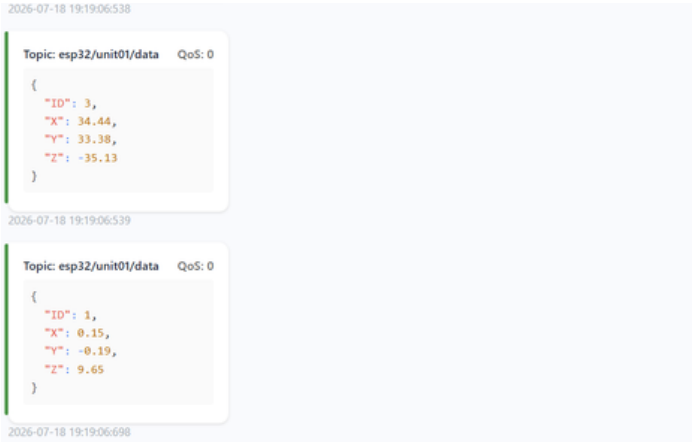
Method

- Architected a FreeRTOS multi-task pipeline on the STM32 with dedicated tasks for **IMU, UART, and status indication,** using message queues for communication under a 20KB RAM constraint
- Manually implemented **SHTP driver** over **I2C**, handling boot sequence, channel setup, and sensor setup enabling **concurrent streaming** from all three BNO085 sensors
- Bridged structured sensor data over **UART** to an ESP32 **MQTT** publisher supporting **OTA firmware** updates

STM32 RTOS Architecture



MQTT Broker Live Output



Results

- Successfully **streamed calibrated real-time data** from all three BNO085 sensors simultaneously
- Resolved multiple low-level debugging challenges including **stack overflow diagnosis** under memory constraints, SHTP channel flooding from concurrent sensor streams and **FreeRTOS heap exhaustion** from task creation
- Achieved **wireless firmware deployment** capability via OTA updates without physical device access

Features

- FreeRTOS multi-task **architecture with message queues** enabling concurrent sensor acquisition and **UART transmission** within a **20KB RAM constraint**
- Manual **SHTP protocol** implementation over **I2C handling** advertisement parsing, Set Feature commands and multi-sensor data report parsing **without external libraries**
- Real-time accelerometer, gyroscope and magnetometer fusion w/Q-point scaling
- **MQTT wireless telemetry pipeline** via ESP32 bridge with OTA firmware update support

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MICROMOUSE - UGA IEEE

International MicroMouse Competition



Situation

- Needed autonomous maze-solving firmware for a 16x16 competition maze with **real-time** wall detection and **dynamic path calculation**
- Required a navigation firmware that could run efficiently within **limited RAM** without blocking time-critical **peripheral communication**

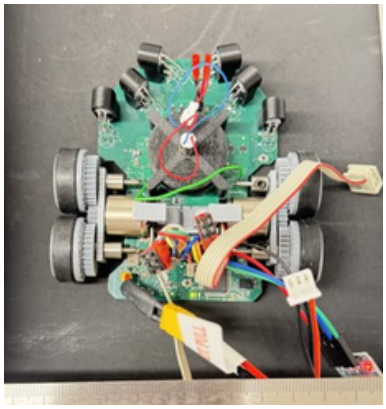
Navigation Feedback Loop Code



Method

- Utilized the **STM32 MCU** as the target hardware to develop all firmware
- Managed concurrent **I2C**, **UART PWM**, and **ADC peripherals** on STM32 using CubeMX and **HAL** drivers for IMU heading, Bluetooth telemetry, motor control, and wall detection
- Integrated all features into a **floodfill navigation firmware** in **embedded C**

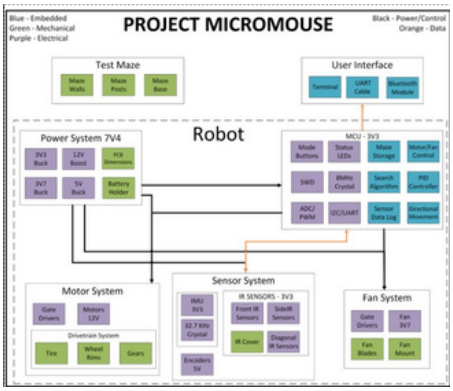
Fully developed MicroMouse robot



Results

- Demonstrated **full maze solve** on test runs before competition, with firmware reliably managing all peripherals simultaneously under real conditions
- Implemented **live Bluetooth telemetry** to validate firmware state during development, confirming correct peripheral timing and navigation decisions in real time

Sensor Communication Diagram



Features

- **Multi-peripheral firmware architecture** coordinating ADC, I2C, UART, and PWM simultaneously on a single STM32
- **Real-time floodfill execution** within RAM and CPU constraints leveraging **DMA**-offloaded sensor reads, allowing for non-blocking simultaneous peripheral communication
- **Wireless debug pipeline** through UART/ Bluetooth for **live firmware state monitoring** during testing